

Speed of wave propagation, $v = f\lambda = (120)(0.25) = 30 \text{ m/s}$

5(a) A region where an electric charge will experience an electric force.

(b) (i) Path: a curve upwards between the plates and straight and at a tangent to the curve beyond the plates

(ii) 1. $F = Ee$

2. $t = L/v$

3. change in momentum = EeL/v

5(c)(i) At the lowest point of graph,

Electric field = gradient of tangent = 0

For point charge, $E = \frac{Q}{4\pi\epsilon r^2}$

Electric field due to charge A = Electric Field due to charge B

$$\begin{aligned}\therefore Q_A / Q_B &= (r_A / r_B)^2 \\ &= (35.0/15.0)^2 \\ &= 5.44\end{aligned}$$

(ii) When $x = 25.0 \text{ cm}$,

$$\text{Gradient, } E = \frac{dV}{dr} = -\frac{540-215}{(2.0-40.0)\times 10^{-2}} = 855 \text{ N C}^{-1} \text{ from A to B}$$

$$F = eE = (1.60 \times 10^{-19})(855) = 1.4 \times 10^{-16} \text{ N}$$

Force on electron is $1.4 \times 10^{-16} \text{ N}$ towards sphere A.